Customer Engagement & Product Utilization Analytics for Retention Strategy

A behavioral churn analysis for The European Central Bank

Prepared for: Unified Mentor | Dataset: European_Bank.csv (10,000 customers)

1. Executive Overview

This paper evaluates customer retention through the lens of behavior and relationship depth rather than demographics alone. Across 10,000 retail banking customers, 20.4% churned. The analysis shows that engagement and product depth explain far more of the variation in churn than balance or salary — high-balance customers are not automatically loyal, and a large share of at-risk value sits with premium, inactive customers who look financially healthy but are quietly disengaging.

2. Methodology

The dataset (CustomerId, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, Exited) was validated for completeness (no missing values, binary fields confirmed as 0/1, churn label confirmed binary). Customers were then classified into engagement profiles, cross-tabulated against product usage and balance tiers, and scored with a composite Relationship Strength Index (RSI).

- Engagement Classification: Active Engaged (Multi-Product), Active but Low-Product, Inactive Disengaged
- Product Utilization Analysis: churn by product count, single- vs multi-product retention
- Financial Commitment vs Engagement: balance tier × activity cross-analysis, at-risk premium detection
- Retention Strength Assessment: RSI construction and churn correlation

3. Key Findings

3.1 Engagement is the strongest behavioral driver of churn

Inactive customers churn at roughly 1.9x the rate of active customers (26.9% vs 14.3%). Customers who are both active and hold 2+ products churn least of all (9.7%), while inactive customers churn most (26.9%).

Figure 1 — Churn rate by engagement profile

3.2 Product depth matters, but only up to a point

Single-product holders churn at 27.7%, while two-product holders drop to just 7.6% — the stickiest segment in the portfolio. Churn then spikes sharply for customers holding 3 products (82.7%) or 4 products (100%), which is far more likely to reflect forced cross-sell, account consolidation ahead of exit, or data-entry edge cases than genuine loyalty, and merits a follow-up review of how those accounts were opened.

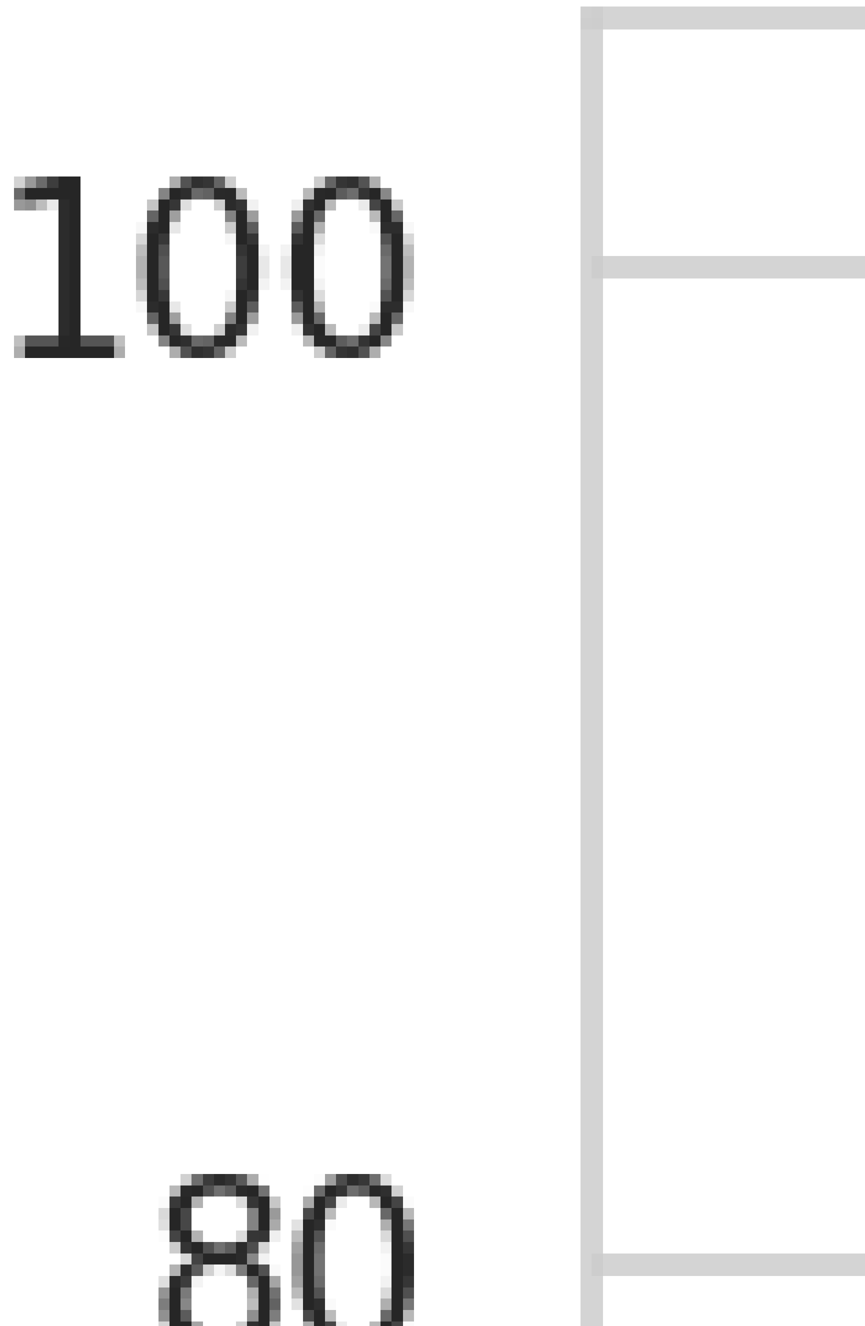
100
80

Figure 2 — Churn rate by number of products held

3.3 High balances do not protect against churn without engagement

Splitting customers into balance quartiles and crossing with activity status shows that inactive customers churn more than active customers in every single balance tier, including the top quartile (30.5% inactive vs 16.9% active). Balance alone is not a retention signal — engagement is.

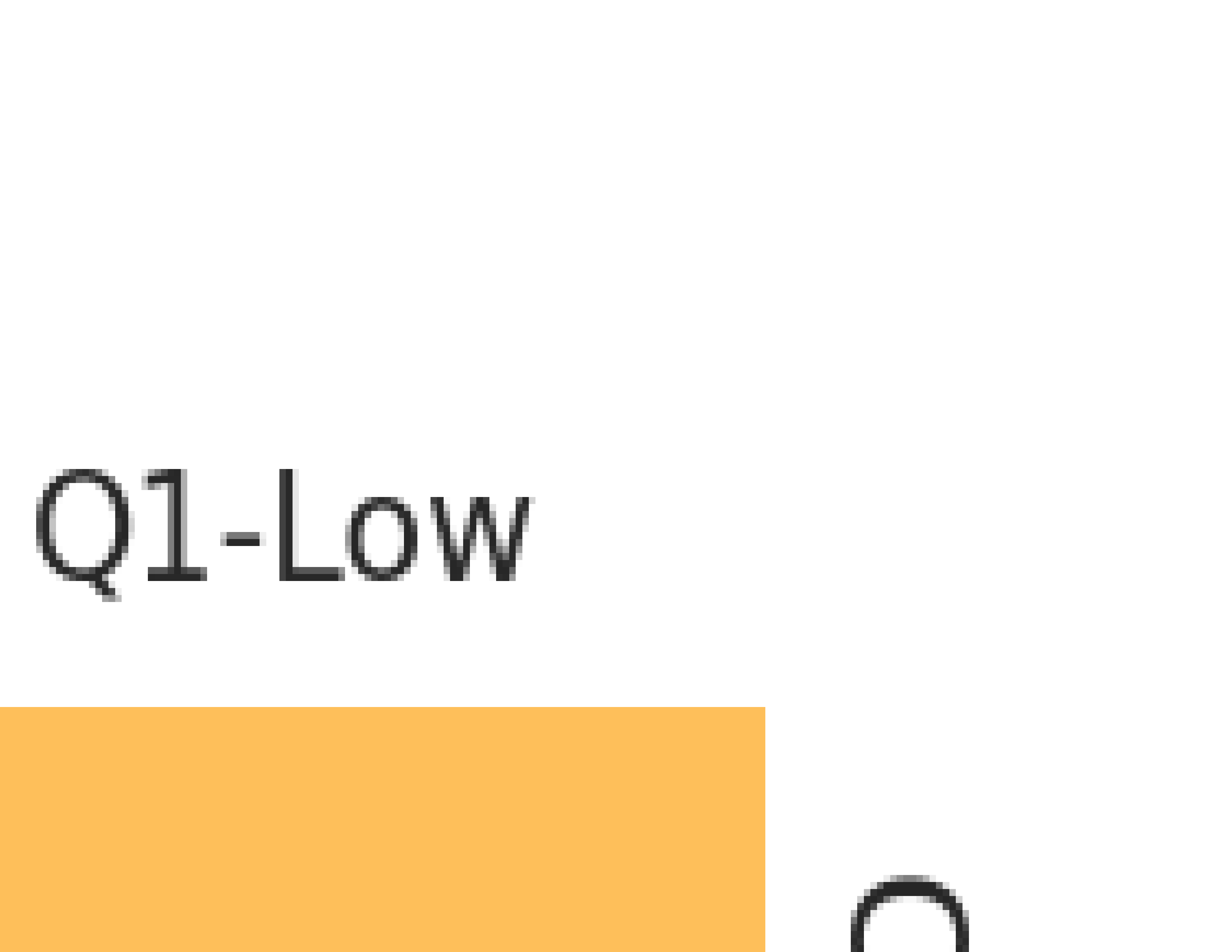
Q1-Low

Figure 3 — Churn rate (%) by balance tier and activity status

3.4 Geography shows a material churn gap

Germany shows a materially higher churn rate (32.4%) than France (16.2%) or Spain (16.7%), suggesting either a market-specific competitive pressure or a service/engagement gap worth investigating regionally.


30

Figure 4 — Churn rate by geography

3.5 Churn risk rises sharply in the 40–60 age band

Churn climbs from 7.5% for customers under 30 to a peak of 56.2% for the 50–60 age band, before falling again for 60+. This mid-to-late-career segment is the highest-priority group for proactive retention outreach.

50

Figure 5 — Churn rate by age group

4. Key Performance Indicators

KPI	Definition	Result
Engagement Retention Ratio	Inactive churn rate ÷ active churn rate	1.88x
Product Depth Index	Churn rate by product count (1→4)	27.7% / 7.6% / 82.7% / 100%
High-Balance Disengagement Rate	Churn rate among top-quartile-balance, inactive customers	30.5%
Credit Card Stickiness Score	Churn-rate gap, no-card minus card-holder	+0.6 pp (weak effect)
Relationship Strength Index	Correlation between composite RSI and churn	-0.15 (moderate, directionally consistent)

5. At-Risk Premium Customers (Silent Churn)

1,247 customers sit in the top balance quartile while being inactive. This group churns at 30.5% — well above the portfolio average of 20.4% — despite representing some of the bank's highest-value relationships. These customers are the clearest target for a proactive, engagement-first retention program, since they are not detectable through balance or salary screening alone.

6. Recommendations

- Prioritize activation over acquisition: re-engagement campaigns for inactive members should outrank new-customer acquisition spend, given the ~1.9x churn multiplier.
- Promote the two-product 'sweet spot' as the target cross-sell outcome, rather than maximizing product count — investigate why 3–4 product accounts show extreme churn before expanding those bundles.
- Build a standing 'at-risk premium' watch-list (top-quartile balance + inactive) for relationship managers, refreshed monthly.
- Investigate Germany-specific churn drivers (pricing, competition, service model) given its outsized churn rate versus France and Spain.
- Target retention outreach at the 40–60 age band, where churn risk is highest and where customers likely have the most product and balance depth to protect.

7. Conclusion

Reframing churn around behavior and relationship depth — rather than demographics or account size — surfaces a clearer, more actionable picture: engagement and the right product mix predict retention far better than wealth indicators alone. A relationship-strength-first retention strategy, anchored on the KPIs above, gives the bank a quantitative basis for prioritizing outreach, redesigning cross-sell, and protecting silently at-risk premium relationships.